



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2011

PRELIMINARY SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 3: FINAL PRELIMINARY EXAMINATION

Mathematics Extension I

TIME ALLOWED: 1½ HOURS

Outcomes Assessed	Questions	Marks
Demonstrates the ability to manipulate and simplify expressions to solve problems involving inequalities, division of an interval into a given ratio and circle geometry.	1	12
Solves problems involving polynomials, angle between two lines, quadratics and graphs.	2	10
Uses appropriate techniques to solve problems involving radians, trigonometric rules and graphs.	3	11
Uses appropriate techniques to solve problems involving trigonometric identities and working in three dimensions.	4	14
Uses appropriate techniques to solve problems involving trigonometric equations.	5	13
Solves problems using differentiation techniques.	6	12

Question	1	2	3	4	5	6	Total	%
Marks	/12	/10	/11	/14	/13	12	/72	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

QUESTION ONE (12 marks)

a) Factorise $8x^3 + 27$

2

b) Solve the inequality

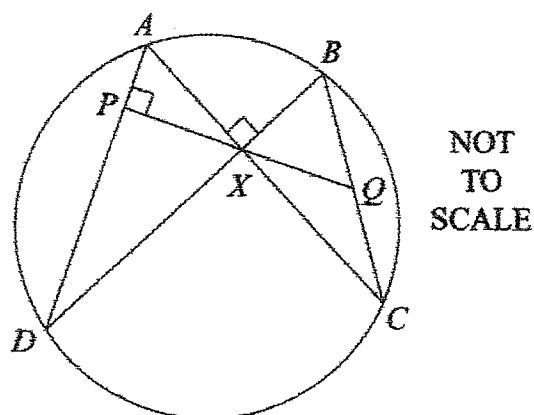
3

$$\frac{x+3}{2x} > 1$$

c) The interval AB, where A is (4,5) and B is (19,-5), is divided internally in the ratio 2:3 by the point P (x,y). Find the values of x and y.

2

d)



The diagram show points A, B, C and D on a circle. The lines AC and BD are perpendicular and intersect at X. The perpendicular to AD through X meets AD at P and BC at Q.

Copy or trace this diagram into your writing booklet.

(i) Prove that angle QXB = angle QBX.

3

(ii) Prove that Q bisects BC.

2

QUESTION TWO (10 marks)

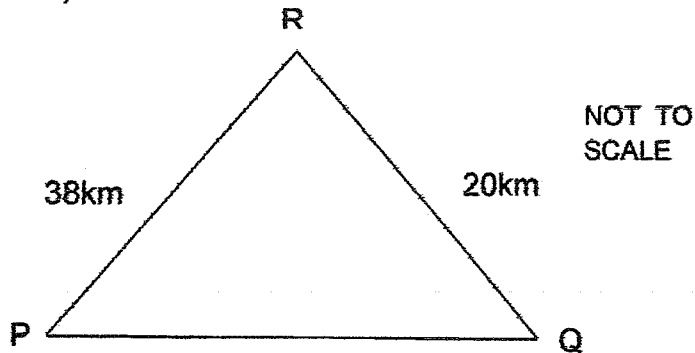
- a) The polynomial $P(x) = x^2 + ax + b$ has a zero at $x=2$.
When $p(x)$ is divided by $x+1$, the remainder is 18. Find the values of a and b . **3**
- b) The cubic polynomial $P(x) = x^3 + rx^2 + sx + t$, where r, s and t are real numbers has three real zeros, 1, α and $-\alpha$.
- (i) Find the value of r . **1**
- (ii) Find the value of $s + t$. **2**
- c) The acute angle between the lines $y = 3x + 5$ and $y = mx + 4$ is 45° . Find two possible values of m . **2**
- d) For what values of k does $x^2 - kx + 4 = 0$ have no real roots? **2**

QUESTION THREE (11 marks)

- a) Find the exact area of an isosceles triangle with equal sides 8cm about an included angle of 45° .

2

b)



In the diagram, the point Q is due east of P.
 The point R is 38km from P and 20km from Q.
 The bearing of R from Q is 325° .

- (i) What is the size of angle PQR?

1

- (ii) What is the bearing of R from P?

3

- c) An arc of length 5 units subtends an angle of θ at the centre of a circle of radius 3 units. Find the value of θ to the nearest degree.

2

- d) (i) On the same set of axes, sketch the graphs of $y = \cos 2x$ and $y = \frac{x+1}{2}$ for $-\pi \leq x \leq \pi$.

2

- (ii) Use your graph to determine how many solutions there are to the equation $2\cos 2x = x + 1$ for $-\pi \leq x \leq \pi$.

1

QUESTION FOUR (14 marks)

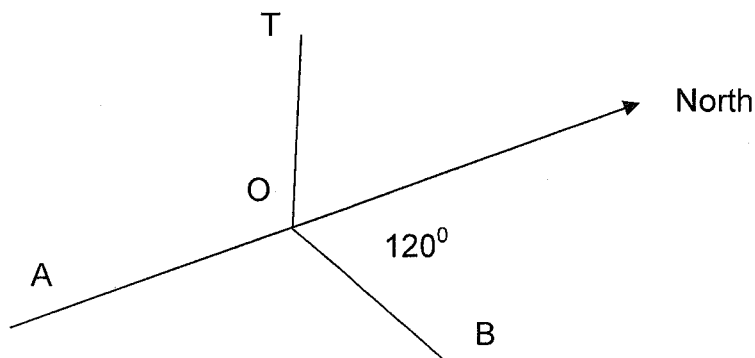
a) Simply fully $\cos 5x \cos 3x + \sin 5x \sin 3x$. 2

b) Given that θ is acute and $\sin 2\theta = \frac{5}{13}$, find $\tan \theta$. 4

c) (i) Prove that $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$
provided that $\cos 2\theta \neq -1$ 2

(ii) Hence show the exact value of
 $\tan \frac{\pi}{8}$ is $\sqrt{2} - 1$ 2

d) From a point A due south of a tower, the angle of elevation of the top of the tower T, is 23° . From another point B, on a bearing of 120° from the tower, the angle of elevation of T is 32° . The distance AB is 200 metres.



(i) Copy or trace the diagram, adding the given information to your diagram. 1

(ii) Hence find the height of the tower to the nearest metre. 3

QUESTION FIVE (13 marks)

- a) (i) Express $\sin x$ and $\cos x$ in terms of t , 2
where $t = \tan \frac{x}{2}$.
- (ii) Use the t formula to solve $\sin x + \cos x = -1$ 4
for $0 \leq x \leq 2\pi$.
- b) (i) Express $3\sin x + 4\cos x$ in the form $A\sin(x + \alpha)$ 2
where $0 \leq \alpha \leq \frac{\pi}{2}$.
- (ii) Hence or otherwise, solve $3\sin x + 4\cos x = 5$ 2
for $0 \leq x \leq 2\pi$. Give your answer or answers,
in radians correct to two decimal places.
- c) Find the general solution for $\sqrt{2}\cos 2\theta + 1 = 0$ 3

QUESTION SIX (12 marks)

a) Find the derivative of each of the following functions:

(i) $y = \frac{1}{x} + \frac{1}{x^2}$ (Answer with positive indices) 2

(ii) $y = \sqrt{x}$ (Answer in surd form) 2

(iii) $y = (3x - 1)(3x^2 + 1)$ 2

(iv) $y = (3x^2 - 2x - 1)^4$ 1

(v) $y = \frac{4x^2}{x^2+5}$ 2

b) (i) Write down the rule for differentiation by first principles. 1

(ii) Hence show by first principles that the derivative of

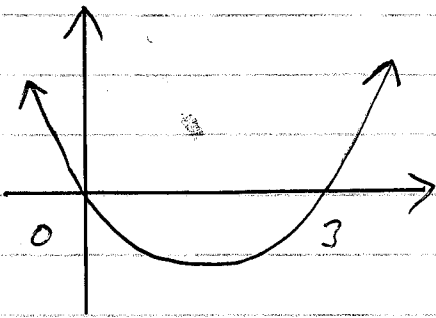
$3x^2 - 4x$ is $6x - 4$. 2

①

YEAR 11 EXTENSION ONEPRELIMINARY COURSE EXAMINATION 2011QUESTION ONE

$$a) \quad 8x^3 + 27 = (2x)^3 + 3^3 = (2x+3)(4x^2 - 6x + 9) \quad \checkmark$$

$$b) \quad \frac{x+3}{2x} > 1 \quad 4x^2 \left(\frac{x+3}{2x} \right) > 4x^2 \quad \checkmark$$



$$2x(x+3) > 4x^2$$

$$2x^2 + 6x > 4x^2$$

$$0 > 2x^2 - 6x$$

$$2x^2 - 6x < 0$$

$$2x(x-3) < 0 \quad \checkmark$$

$$\text{zeros } x=0 \quad x=3$$

$$\therefore 0 < x < 3 \text{ from graph} \quad \checkmark$$

$$c) \quad A(4, 5) \quad B(19, -5) \quad P(x, y)$$

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n} \quad m:n = 2:3$$

$$x = \frac{2 \times 19 + 3 \times 4}{2+3}$$

$$y = \frac{2 \times -5 + 3 \times 5}{2+3} \quad \checkmark$$

$$= \frac{50}{5}$$

$$= \frac{5}{5}$$

$$\therefore x = 10$$

$$y = 1 \quad \checkmark$$

(3)

QUESTION TWO

(c) $P(x) = x^2 + ax + b$

Zero at $x=2 \quad \therefore P(2) = (2)^2 + a(2) + b = 0$
 $4 + 2a + b = 0 \quad (1)$

$P(-1) = 18$, by remainder theorem
 $(-1)^2 + a(-1) + b = 18$ ✓

$1 - a + b = 18 \quad (2)$

From (1)

$2a + b = -4 \quad (3)$

From (2)

$-a + b = 17 \quad (4)$

(3) - (4)

$3a = -21$

$a = -7$ ✓

Sub $a = -7$ in (1) $4 + 2 \times -7 + b = 0$

$-10 + b = 0$

$b = 10$ ✓

$\therefore a = -7 \quad b = 10$

(b) (i) $P(x) = x^3 + rx^2 + sx + t$

The sum of zeros $= -\frac{b}{a}$

$1 + \alpha + (-\alpha) = -\frac{r}{1}$

$r = -1$ ✓

(ii) The product of zeros $= -\frac{d}{a}$

$1 \times \alpha \times (-\alpha) = -\frac{t}{1}$

$\alpha^2 = t$ ✓

The sum of zeros 2 at a time $= \frac{s}{a}$

$1\alpha + \alpha(-\alpha) + (\alpha)1 = \frac{s}{1}$

$\alpha - \alpha^2 + \alpha = s$

$s = -\alpha^2$

$s + t = -\alpha^2 + \alpha^2 = 0$ ✓

(4)

(c) $y = 3x + 5$ has $m_1 = 3$
 $y = mx + 4$ has $m_2 = m$

If θ is the angle between the lines

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad \text{and} \quad \theta = 45^\circ$$

$$\tan 45^\circ = \left| \frac{3 - m}{1 + 3m} \right| \quad \checkmark$$

$$\frac{3 - m}{1 + 3m} = 1 \quad \text{or}$$

$$3 - m = 1 + 3m$$

$$2 = 4m$$

$$m = \frac{1}{2}$$

$$\frac{3 - m}{1 + 3m} = -1$$

$$3 - m = -1 - 3m$$

$$2m = -4$$

$$m = -2$$

$$\therefore m = \frac{1}{2} \text{ or } -2 \quad \checkmark$$

(d) $x^2 - kx + 4 = 0$

$$\Delta = b^2 - 4ac$$

$$= (k)^2 - 4 \times 1 \times 4$$

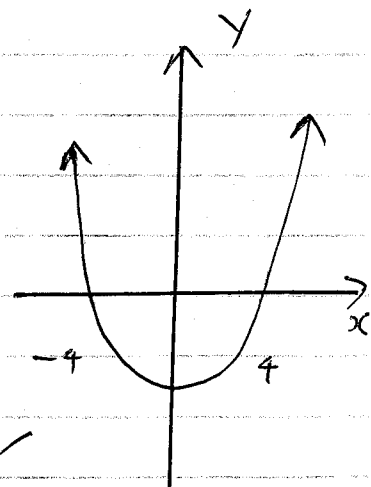
$$= k^2 - 16$$

For no real roots $\Delta < 0$

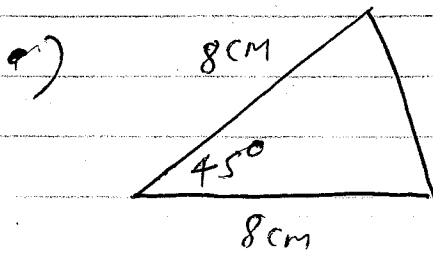
$$k^2 - 16 < 0 \quad \checkmark$$

$$(k - 4)(k + 4) < 0$$

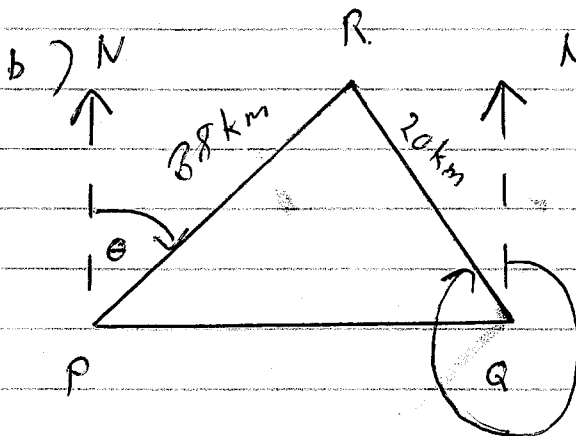
$$-4 < k < 4 \quad (\text{from graph})$$



(5)

QUESTION THREE

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} \times 8 \times 8 \times \sin 45^\circ \\
 &= \frac{1}{2} \times 8 \times 8 \times \frac{1}{\sqrt{2}} \quad \checkmark \\
 &= \frac{32}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\
 &= 16\sqrt{2} \text{ cm}^2. \quad \checkmark
 \end{aligned}$$



$$\begin{aligned}
 \text{(i)} \quad \angle PQR &= 325^\circ - 270^\circ \\
 &= 55^\circ \quad \checkmark
 \end{aligned}$$

(ii) Let bearing equal θ as shown.

$$\frac{\sin \angle RPQ}{20} = \frac{\sin 55^\circ}{38} \quad \checkmark$$

$$\sin \angle RPQ = \frac{20 \sin 55^\circ}{38}$$

$$\sin \angle RPQ = 0.421132654 \dots$$

$$\angle RPQ = 25^\circ 32' \quad \checkmark$$

$$\begin{aligned}
 \text{Bearing of } R \text{ from } P &= 90^\circ - 25^\circ 32' \\
 &= 64^\circ 28' \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 &= 064^\circ \text{ (correct to nearest degree)} \quad \checkmark
 \end{aligned}$$

6

(c) $l = r \theta$ (θ in radians)

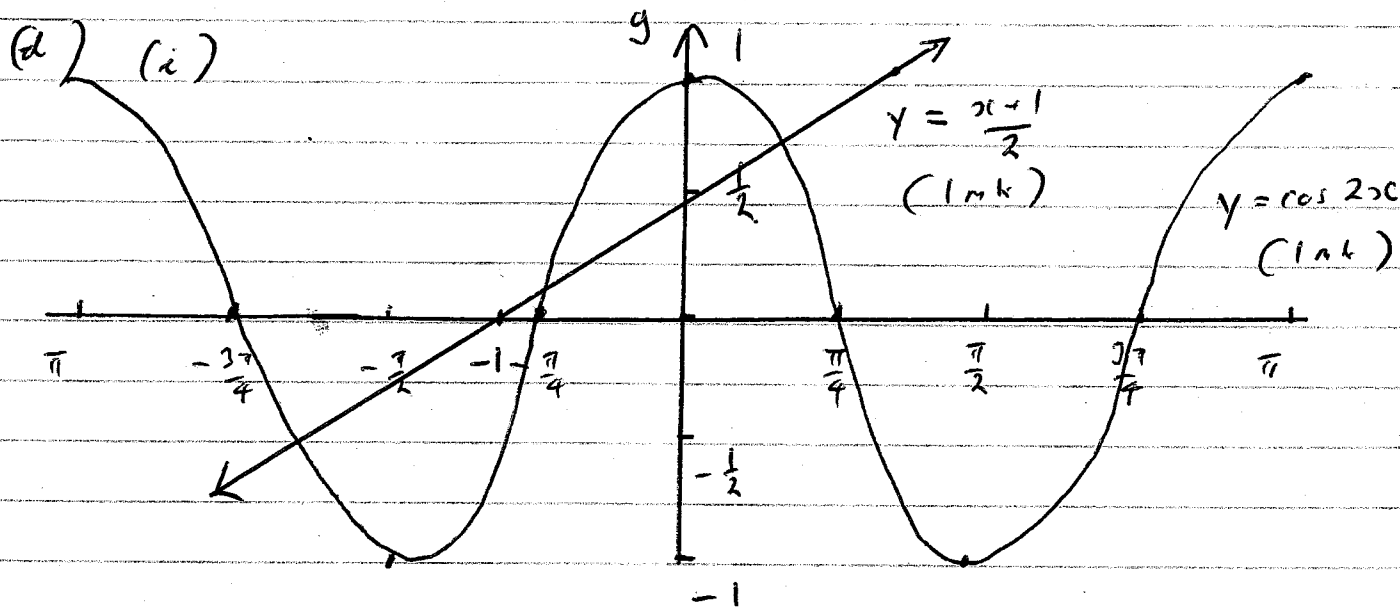
$5 = 3\theta$ ✓

$\theta = \frac{5}{3}$

$= \frac{5}{3} \times \frac{180}{\pi}$

$= 95.49296586...$

$= 95^\circ$ (nearest degree) ✓

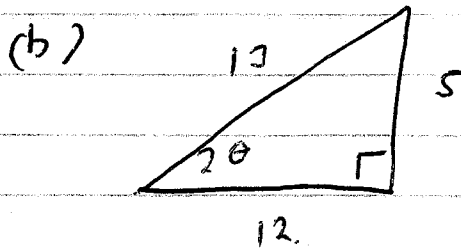


(ii) There are 3 solutions to the equation $2 \cos x = x+1$ for $-\pi \leq x \leq \pi$ (Since the two graphs intersect 3 times in the given domain) ✓

⑦

QUESTION FOUR

(a) $\cos 5x \cos 3x + \sin 5x \sin 3x = \cos (5x - 3x) \checkmark$
 $= \cos 2x \checkmark$



$$\sin 2\theta = \frac{5}{13}$$

$$\tan 2\theta = \frac{5}{12} \checkmark$$

$$\tan (\theta + \theta) = \frac{5}{12}$$

$$\frac{\tan \theta + \tan \theta}{1 - \tan \theta \tan \theta} = \frac{5}{12}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{5}{12} \checkmark$$

$$24 \tan \theta = 5 - 5 \tan^2 \theta$$

$$5 \tan^2 \theta + 24 \tan \theta - 5 = 0$$

$$(5 \tan \theta - 1)(\tan \theta + 5) = 0 \checkmark$$

$$\therefore \tan \theta = \frac{1}{5} \text{ or } \tan \theta = -5$$

but since θ is acute (1st quadrant)

$$\text{then } \tan \theta = \frac{1}{5} \checkmark$$

(c) (i) RHS = $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)} \checkmark$

$$= \frac{2\sin^2 \theta}{2\cos^2 \theta}$$

$$= \tan^2 \theta = \text{LHS} \checkmark$$

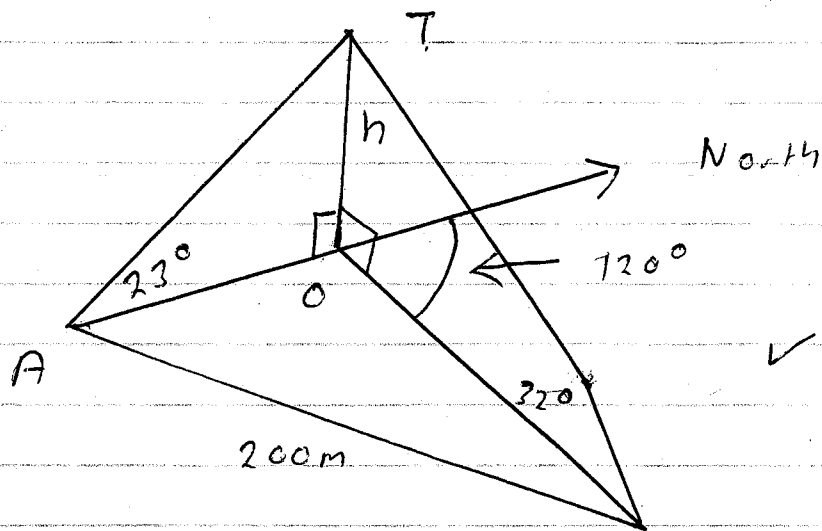
(ii) Let $\theta = \frac{\pi}{8}$ $\tan^2 \frac{\pi}{8} = \frac{1 - \cos \frac{\pi}{4}}{1 + \cos \frac{\pi}{4}} = \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \checkmark$

$$= \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}$$

8

$$\begin{aligned} &= (\sqrt{2}-1)^2 \\ \therefore \tan \frac{\pi}{8} &= \sqrt{2}-1 \quad \checkmark \end{aligned}$$

(d) (i)



(ii) Let $TO = h$ metres In $\triangle AOT$ $\tan 23^\circ = \frac{h}{AO}$
 $\therefore AO = \frac{h}{\tan 23^\circ}$

In $\triangle BOT$ $\tan 32^\circ = \frac{h}{BO}$
 $BO = \frac{h}{\tan 32^\circ}$

$\angle AOB = 180^\circ - 120^\circ$ (from diagram)

$$AB^2 = AO^2 + BO^2 - 2 \times AO \times BO \times \cos 60^\circ$$

$$200^2 = \left(\frac{h}{\tan 23^\circ} \right)^2 + \left(\frac{h}{\tan 32^\circ} \right)^2 - 2 \times \frac{h}{\tan 23^\circ} \times \frac{h}{\tan 32^\circ} \times \cos 60^\circ \quad \checkmark$$

$$= \frac{h^2}{\tan^2 23^\circ} + \frac{h^2}{\tan^2 32^\circ} - \frac{2h^2}{\tan 23^\circ \tan 32^\circ} \times \frac{1}{2}$$

$$h^2 \left(\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 32^\circ} - \frac{1}{\tan 23^\circ \tan 32^\circ} \right) = 40000$$

$$h^2 = \frac{40000}{\left(\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 32^\circ} - \frac{1}{\tan 23^\circ \tan 32^\circ} \right)} \quad \checkmark$$

$$\left(\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 32^\circ} - \frac{1}{\tan 23^\circ \tan 32^\circ} \right)$$

9

$$h = \sqrt{\frac{40000}{4.349 \dots}}$$

$$= 95.9924 \dots \text{ m}$$

$$= 96 \text{ m (to nearest m)} \quad \checkmark$$

$\therefore$ Height of the tower is 96m

(10)

QUESTION FIVE

(a) (i) $\sin x = \frac{2t}{1+t^2}$ ✓ $\cos x = \frac{1-t^2}{1+t^2}$ ✓

(ii) $\sin x + \cos x = -1$
 $\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = -1$ ✓

$$2t + 1 - t^2 = -1 - t^2$$

$$2t = -2$$

$$t = -1$$

$$\therefore \tan \frac{x}{2} = -1$$
 ✓

$$\frac{x}{2} = 135^\circ, 215^\circ$$

$$x = 270^\circ, 430^\circ$$

$$x = \frac{3\pi}{2} \text{ for } 0 \leq x \leq 2\pi$$
 ✓

Final check sub $x = \pi$ in original eqn.

$$\sin \pi + \cos \pi = -1 \quad \text{TRUE}$$

$$\therefore x = \pi, \frac{3\pi}{2} \text{ for } 0 \leq x \leq 2\pi$$
 ✓

(11)

$$(b) \quad (i) \quad 3 \sin x + 4 \cos x = A \sin(x + \alpha)$$

$$A = \sqrt{4^2 + 3^2} = 5 \quad \alpha = \tan^{-1}\left(\frac{4}{3}\right)$$

$$3 \sin x + 4 \cos x = 5 \sin\left(x + \tan^{-1}\left(\frac{4}{3}\right)\right)$$

$$(ii) \quad 5 \sin\left[x + \tan^{-1}\left(\frac{4}{3}\right)\right] = 5$$

$$\sin\left[x + \tan^{-1}\left(\frac{4}{3}\right)\right] = 1$$

$$x + \tan^{-1}\left(\frac{4}{3}\right) = \frac{\pi}{2} \quad \checkmark$$

$$\text{or } x = \frac{\pi}{2} - \tan^{-1}\left(\frac{4}{3}\right)$$

$$= \frac{\pi}{2} - 0.9271 \dots$$

$$= 0.64 \text{ (to 2 decimal places)} \quad \checkmark$$

$$(c) \quad \sqrt{2} \cos 2\theta + 1 = 0$$

$$\cos 2\theta = -\frac{1}{\sqrt{2}} \quad \checkmark$$

$$2\theta = 2n\pi \pm \frac{3\pi}{4} \quad \checkmark$$

$$\theta = n\pi \pm \frac{3\pi}{8} \quad \checkmark$$

(12)

QUESTION SIX

$$a) \quad i) \quad y = \frac{1}{x} + \frac{1}{x^2}$$

$$= x^{-1} + x^{-2}$$

$$y' = -x^{-2} - 2x^{-3} \quad \checkmark$$

$$= -\frac{1}{x^2} - \frac{2}{x^3} \quad \checkmark$$

$$ii) \quad y = \sqrt{x}$$

$$= x^{\frac{1}{2}}$$

$$y' = \frac{1}{2} x^{-\frac{1}{2}} \quad \checkmark$$

$$= \frac{1}{2\sqrt{x}} \quad \checkmark$$

$$iii) \quad y = (3x-1)(3x^2+1) = (3x-1)6x + (3x^2+1)3 \quad \checkmark$$

$$= 18x^2 - 6x + 9x^2 + 3$$

$$= 27x^2 - 6x + 3 \quad \checkmark$$

$$iv) \quad y = (3x^2 - 2x - 1)^4 = 4(3x^2 - 2x - 1)^3 \times (6x - 2)$$

$$= 8(3x-1)(3x^2 - 2x - 1)^3 \quad \checkmark$$

$$v) \quad y = \frac{4x^2}{x^2+5} = \frac{(x^2+5)8x - 4x^2 \times 2x}{(x^2+5)^2} \quad \checkmark$$

$$= \frac{8x^2 + 40x - 8x^2}{(x^2+5)^2} \quad \checkmark$$

$$= \frac{40x}{(x^2+5)^2}$$

(13)

$$(b) \quad (i) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \checkmark$$

$$(ii) \quad f'(x) = \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 4(x+h)] - [3x^2 - 4x]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{3x^2} + 6xh + 3h^2 - \cancel{4x} - 4h - \cancel{3x^2} + \cancel{4x}}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{6xh - 4h + 3h^2}{h}$$

$$= \lim_{h \rightarrow 0} 6x - 4 + 3h$$

$$= 6x - 4 \quad \checkmark$$